ABC Trading Java Rewrite

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Implementation Status Report

Scope: `java/src/main/java`, `java/src/test/java`, the Python JPype facade, and the Rust-vs-Java reconciliation tools.

Purpose: describe what is implemented in Java so far, how the pieces fit together, how each class is used, and why the boundaries were chosen.

Status: implementation snapshot as of 2026-08-31.

1. Executive Summary

The Java rewrite is a deterministic, synchronous, backtest-oriented trading runtime shaped after Nautilus Trader. Java owns the runtime loop, message routing, order state, risk boundary, simulated execution, portfolio position state, and event logging. Python owns the strategy callback and historical-data loading at the current integration boundary.

The strongest verified feature is cross-backend behavioral reconciliation. The shared historical-bar workflow currently compares 228 semantic lifecycle rows between Nautilus and Java. The dedicated L3 MBO fixture compares two fills across a named venue order, queue-ahead trades, maker/taker classification, and deterministic fill ordering.

Verified results

```
mvn -q clean test
MATCH rows=228
MATCH L3 fills=2
MATCH account state fields=8
MONETARY exact API pass
```

The current Java test suite covers the deterministic runtime, L3 queue behavior, and account settlement. Account-state events expose total, locked, free, initial-margin, maintenance-margin, and currency information.

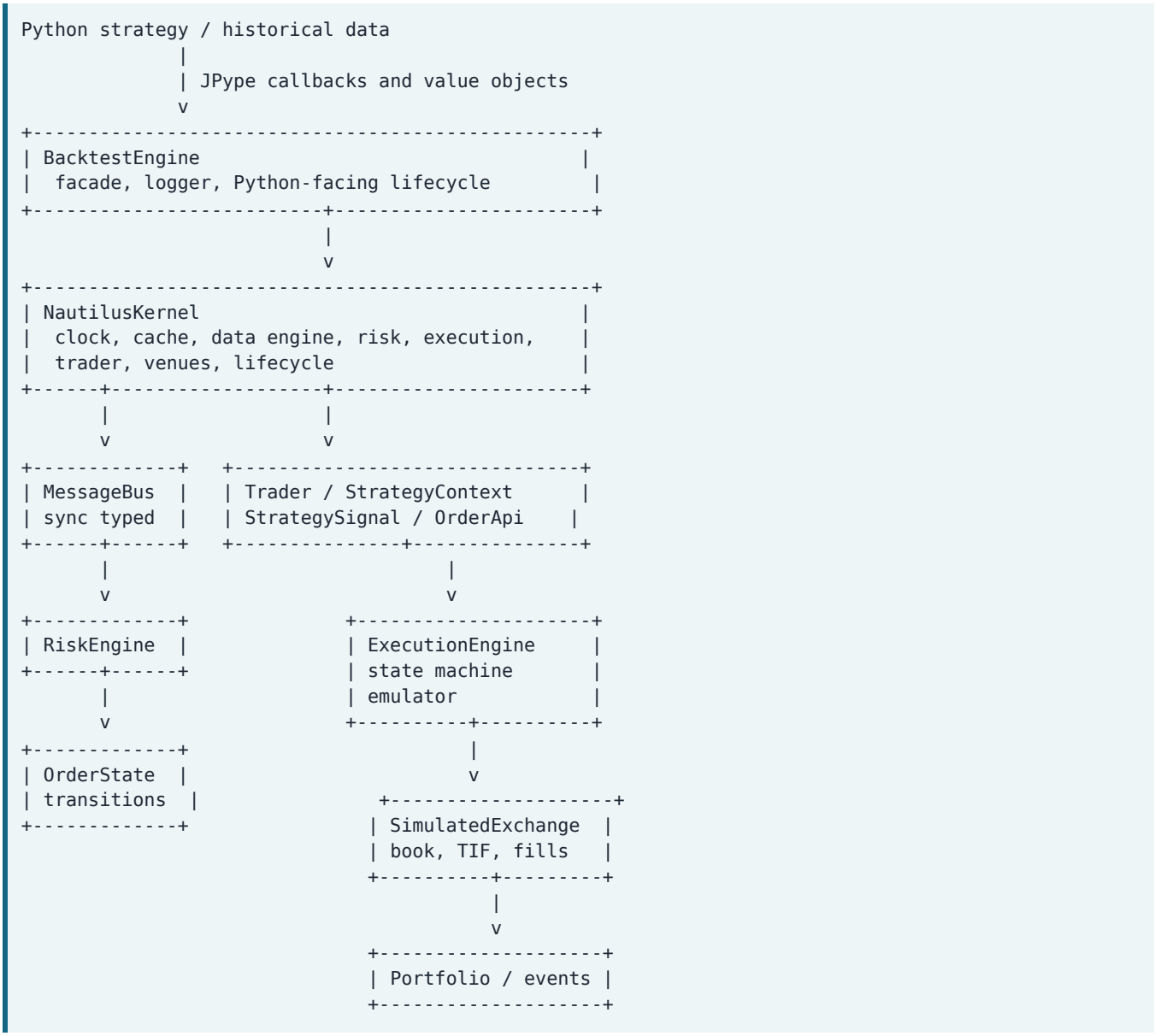
Main implemented feature groups

- synchronous typed message bus with topic routing, priorities, wildcards, endpoints, and correlation callbacks
- component lifecycle and kernel composition
- Python strategy callbacks through JPype
- market and limit orders
- stop-market and stop-limit orders
- trailing stop market and trailing stop limit orders
- order lifecycle state machine
- cancel, modify, reject, deny, expiry, IOC, FOK, GTD, and DAY behavior
- local order emulation
- fixed tick-size configuration for Rust-supported `TICKS` trailing offsets
- L2 aggregate order-book snapshots and deltas

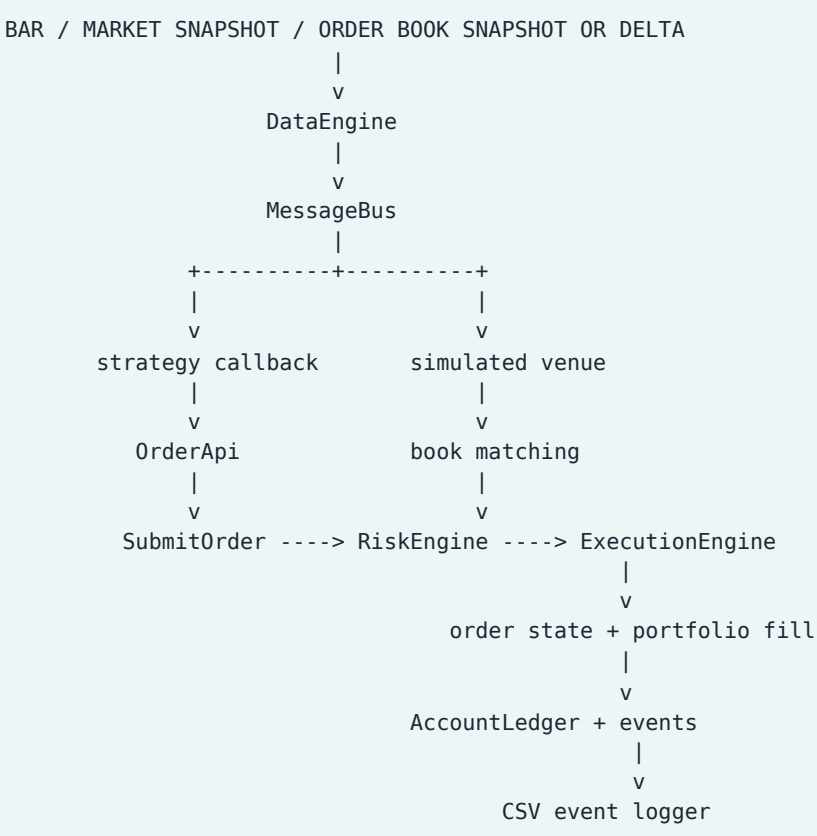
- L3 individual venue-order snapshots and deltas
- trade-driven L3 queue-ahead consumption
- partial fills and multiple fills across price levels
- maker/taker liquidity classification
- static operation latency
- multiple fee models
- cash and margin account settlement
- instrument-specific margin rates and explicit FX conversion
- multi-currency balance views and account-state events
- CSV lifecycle logging
- Rust-vs-Java reconciliation fixtures and comparators

2. Architecture

2.1 Runtime composition



2.2 Data and order flow



2.3 Design principles

1. **Synchronous by default.** The core `MessageBus` calls handlers during `publish`; it is not a worker queue.
2. **Explicit boundaries.** Data, risk, execution, portfolio, strategy, and reconciliation are separate Java packages.
3. **Deterministic ordering.** Input bars are sorted by timestamp and symbol. Bus handlers use priority and registration order. Working orders use price priority and insertion sequence.
4. **Immutable messages.** Records are used for commands, events, market snapshots, book levels, and order state.
5. **Exact quantities at the contract boundary.** Quantities use immutable fixed-point `Quantity` values backed by `BigDecimal` raw values and explicit precision. Prices and PnL remain primitive-backed `double` values for now.
6. **Source anchoring.** Rust paths are recorded in capability metadata and the reconciliation harness compares observable behavior rather than only totals.

3. Package Structure

```
com.abc.trading
|-- backtest          runner facade, iterator, capability metadata
|-- cache             instruments, positions, orders
|-- common.factories  typed order construction
|-- data              bars, market snapshots, book snapshots/deltas, ticks
|-- events            event schema and CSV logger
|-- execution         commands, state machine, venue, fills, fees, latency
|-- indicators        EMA and SMA
|-- model             money and immutable order models
|-- msgbus            typed synchronous bus and transport backings
|-- portfolio         positions, balances, margin, PnL, account events
|-- reconciliation    Java-side comparison DTOs
|-- risk              risk decisions and trading state
|-- system            kernel, clock, trader, component lifecycle
`-- trading           actors, strategies, context, order API
```

The Python bridge lives under `python/abc_trading`; reconciliation scripts live under `recon`.

4. Feature Matrix

Feature	Status	Implementation surface	Notes
Typed synchronous message bus	Implemented	`msgbus.MessageBus`, `TypedTopicRouter`, `TypedEndpointMap`	Direct dispatch, wildcard topics, priorities, endpoint sends, request/response correlation
Kernel lifecycle	Implemented	`NautilusKernel`, `ComponentLifecycle`	Initialization, start, stop, reset, dispose, fault/degrade vocabulary
Strategy callbacks	Implemented	`Trader`, `StrategyContext`, Python `BacktestEngine`	Python strategy receives Java-driven bars and context
Market orders	Implemented	`OrderApi`, `ExecutionEngine`, `SimulatedExchange`	Immediate or latency-delayed execution
Limit orders	Implemented	same	Cross book levels or rest as maker
Stop market/limit	Implemented	order models, trigger evaluation, state machine	Stop-market fills immediately; stop-limit enters `TRIGGERED`
Trailing stops	Implemented	trailing models, exchange ratchet	`PRICE`, `BASIS_POINTS`, `TICKS`; `PRICE_TIER` intentionally rejected to match current Rust
Cancel/modify	Implemented	command records, execution client, state machine	Explicit pending and reject states
TIF	Implemented	`TimeInForce`, exchange	`GTC`, `IOC`, `FOK`, `GTD`, `DAY`
Partial/multiple fills	Implemented	working orders and `OrderStateMachine`	Quantity and average fill price tracked
L2 book snapshots	Implemented	`OrderBookSnapshot`, `BookLevel`	Aggregate depth per price level
L2 book deltas	Implemented	`OrderBookDelta`, `BookAction`	`ADD`, `UPDATE`, `DELETE`, `CLEAR`

Feature	Status	Implementation surface	Notes
L3 individual venue queue	Implemented	`VenueOrder`, `OrderBookL3Snapshot`, `OrderBookL3Delta`, `TradeTick`, `SimulatedExchange`	Individual order identity, price-time priority, queue-ahead deltas, and trade-driven queue consumption
Liquidity classification	Implemented	`OrderFill.liquiditySide`	`MAKER`/`TAKER` included in reconciliation CSV
Fees	Implemented	`FeeModel` implementations	Fixed, maker/taker, per-contract, probability, capped, notional/tiered
Static latency	Implemented	`LatencyModel`, `StaticLatencyModel`	Operation latency with deterministic timestamp ordering
Position/PnL accounting	Implemented	`Portfolio`, `PositionUpdate`, `Money`	Net position, average price, realized PnL, and commission effects
Cash/margin account	Implemented baseline	`AccountLedger`, `AccountState`, `AccountBalance`, `AccountType`	Reservations, free/locked balances, cash settlement, initial/maintenance margin, and threshold events
Instrument price and size metadata	Implemented	`InstrumentSpec`, `Cache`	Quote/base currencies, margin rates, `sizePrecision`, `sizeIncrement`, `pricePrecision`, and `priceTickSize`
FX conversion	Implemented	`FxRateUpdate`, `AccountLedger`, `Portfolio`, `BacktestEngine`	Replayable market-data rates for cross-currency margin and PnL conversion
Account-state events	Implemented	`AccountStateEvent`, `Event`, `CsvEventLogger`	Balance and margin transitions are observable in canonical logs
Mark-to-market valuation	Implemented baseline	`Portfolio`, `AccountLedger`, `MarketDataSnapshot`	Mark updates recalculate signed unrealized PnL and threshold flags
Margin model	Implemented baseline	`MarginModelType`, `InstrumentSpec`, `AccountLedger`	Notional-rate and fixed-per-unit formulas with leverage
Decimal accounting	Implemented at monetary boundaries	`Money`, `Commission`, `AccountBalance`, `AccountState`, `AccountLedger`	Exact `BigDecimal` balances, commissions, FX, reservations, margins, equity, and Python `Decimal` reporting; legacy double projections remain for compatibility
Local order emulator	Implemented	`OrderEmulator`	Snapshot-triggered local ownership and release
Disruptor bus	Scaffold only	`DisruptorMessageBus`	Publish method is still a TODO
External ring-buffer backing	Implemented as backing	`RingBufferMessageBusBacking`	Bounded queue; full-buffer policy currently drops and reports
Persistence/event store	Implemented	`PersistentEventStore`, `EventReplayer`, `EventCheckpoint`	Versioned append-only JSONL, projections, synchronous replay, and checkpoint resume

Feature	Status	Implementation surface	Notes
Binance USD-M Futures adapter	Implemented baseline	`BinanceFuturesAdapter`, `BinanceFuturesLiveRuntime`	Public streams, signed REST, user data, reconnects, and kernel routing; Testnet credentials remain opt-in
Live adapter precision	Implemented	`BinanceInstrumentMetadata`, `BinanceOrderValidator`, `Quantity`	Binance `stepSize` and `tickSize` derive exact size/price rules; invalid values are rejected before REST

5. Class and Type Catalog

The following catalog covers the Java production types currently under `java/src/main/java`.

5.1 Backtest package

Type	Usage	Design rationale
`BacktestEngine`	Public JPype-friendly facade; registers venues/instruments/strategies and runs data	Keeps Python integration simple while Java owns orchestration and logging
`BacktestDataIterator`	Stores and iterates historical bars	Mirrors a deterministic replay boundary and supports reset/clear
`BacktestEngineConfig`	Immutable runner options	Makes sort/stream/analysis intent explicit even where the current facade uses defaults
`BacktestResult`	Immutable run summary DTO	Separates result reporting from runtime state
`EngineCapability`	One capability/status record	Makes Rust mapping and implementation status inspectable
`EngineCapabilities`	Registry of current capability mappings	Documents implemented and pending Rust-aligned surfaces
`SimulatedVenueConfig`	Venue flags, latency, and fee configuration	Keeps venue behavior configurable without changing kernel composition

5.2 Cache and common factory packages

Type	Usage	Design rationale
`Cache`	Instrument venue mapping, positions, and recorded orders	Central deterministic state store shared by risk, execution, and portfolio
`OrderFactory`	Creates client IDs and typed market/limit/stop/trailing orders	Centralizes identity generation and model construction

5.3 Data package

Type	Usage	Design rationale
`Bar`	Immutable close-price input for strategy and legacy replay	Minimal bar boundary with primitive fields and stable sequence
`MarketDataSnapshot`	Carries bid, ask, last, mark, index	Gives trigger evaluation one deterministic multi-source input
`BookLevel`	Price and aggregate quantity at one level	Small immutable unit for L2 depth

Type	Usage	Design rationale
`OrderBookSnapshot`	Complete immutable bid/ask ladder	Snapshot replacement is simple and deterministic for replay
`OrderBookDelta`	One book mutation	Mirrors Rust add/update/delete/clear event semantics
`VenueOrder`	Individual venue order with side, price, size, and sequence	Preserves L3 MBO identity instead of flattening depth into aggregate levels
`OrderBookL3Snapshot`	Complete individual-order bid/ask state	Establishes deterministic price-time priority at snapshot boundaries
`OrderBookL3Delta`	Add, update, delete, or clear one venue order	Allows queue position to respond to exact venue-order mutations
`TradeTick`	Executed market trade with aggressor side	Drives queue-ahead consumption independently from book-depth updates
`AggressorSide`	Buyer, seller, or no-aggressor trade classification	Determines which passive queue can advance
`FxRateUpdate`	Replayable FX conversion-rate input	Makes cross-currency account valuation deterministic and time-ordered
`MarginModelType`	Notional-rate or fixed-per-unit formula selection	Keeps instrument margin policy explicit rather than hidden in risk code
`InstrumentSpec`	Base/quote currencies, margin rates, size precision/increment, and price precision/tick size	Carries the instrument facts needed by accounting, risk, and exact order validation
`BookAction`	`ADD`, `UPDATE`, `DELETE`, `CLEAR`	Makes book mutation intent explicit
`TickScheme`	Instrument-owned fixed tick increment	Rust currently supports `TICKS` through `price_increment`; Java avoids a hard-coded exchange tick in that path
`DataEngine`	Publishes bars, market snapshots, books, and deltas	Separates data distribution from data ownership
`DataClient`	Contract for data clients	Extension point for live or external data sources
`DataEngineConfig`	Immutable data-engine options	Configuration boundary for future subscription/replay policies

5.4a Binance adapter package

Type	Usage	Design rationale
`BinanceEnvironment`	Selects live, Testnet, or demo routes	Keeps endpoint policy explicit and aligned with Nautilus URL helpers
`BinanceFuturesConfig`	Symbols, credentials, timeouts, reconnect, GTD, and startup policy	Mirrors Nautilus `BinanceDataClientConfig`/`BinanceExecClientConfig` without embedding secrets
`BinanceHmacSigner`	Signs Binance query strings with HMAC-SHA256	Matches Binance signed REST and Nautilus account-client behavior
`BinanceQuery`	Encodes ordered REST parameters	Keeps signature input deterministic and URL-safe
`BinanceHttpTransport`	Injectable HTTP boundary	Enables offline contract tests and alternate transport implementations
`JavaBinanceHttpTransport`	JDK `HttpClient` implementation	Provides a dependency-light production REST transport

Type	Usage	Design rationale
`BinanceMessageMapper`	Parses public and user WebSocket JSON	Separates wire schema from runtime behavior and preserves Rust-shaped event semantics
`BinancePriceLevel`	Decimal price/quantity pair	Prevents precision loss before a caller chooses a core representation
`BinanceDepthUpdate`	Binance `depthUpdate` record	Preserves update IDs needed for order-book sequencing and gap detection
`BinanceTradeEvent`	Binance `aggTrade` record	Maps `buyerIsMaker` into Nautilus buyer/seller aggressor semantics
`BinanceMarkPriceEvent`	Binance mark/index price record	Supplies futures valuation and trigger inputs
`BinanceOrderUpdate`	`ORDER_TRADE_UPDATE` projection	Carries execution type, status, fill, commission, and reduce-only state
`BinanceAccountUpdate`	`ACCOUNT_UPDATE` projection	Carries wallet, cross-wallet, and per-position unrealized values
`BinanceInstrumentMetadata`	Parsed `exchangeInfo` symbol metadata	Maps filters, currencies, tick size, and margin percentages to `InstrumentSpec`
`BinanceAccountSnapshot`	Parsed signed account response	Maps wallet, available, margin, and unrealized fields to `AccountState`
`BinanceMarketDataHandler`	Public event callback contract	Keeps market data independently testable and routable
`BinanceExecutionHandler`	User-data callback contract	Keeps execution/account events separate from public market data
`BinanceFuturesAdapter`	REST/WebSocket Binance USD-M client	Owns endpoint lifecycle, signing, listen-key renewal, reconnect, and order mapping
`BinanceFuturesLiveRuntime`	Bridges adapter events into Java core records and bus	Makes one real adapter exercise the same risk, execution, portfolio, and event paths as backtest

5.4 Events package

Type	Usage	Design rationale
`Event`	Canonical CSV row with lifecycle, order, PnL, commission, liquidity, balance, and margin	Quantity, position, and monetary fields are exact decimal values at the schema boundary
`EventType`	Semantic event vocabulary	Keeps logger output independent from concrete Java event class names
`EventLogger`	Event logging contract	Allows CSV or another sink later
`CsvEventLogger`	Writes synchronized CSV rows	Human-readable evidence and simple reconciliation input
`EventStoreRecord`	Versioned JSONL envelope with offset and canonical event	Makes persistence schema and append position explicit; decimal values serialize as strings
`PersistentEventStore`	Append-only JSONL event sink and reader	Provides durable audit evidence without coupling the runtime to a database
`CompositeEventLogger`	Fans one event to CSV and persistent sinks	Preserves existing CSV output while adding durable storage
`EventCheckpoint`	Stores next offset and sequence watermarks	Supports restart/resume without replaying downstream notifications twice
`EventReplayState`	Rebuilds orders, positions, PnL, and accounts	Provides a deterministic projection for recovery and audit

Type	Usage	Design rationale
`ReplayOrderState`	Immutable reconstructed order state	Keeps replay output independent from mutable live order objects
`ReplayAccountState`	Immutable reconstructed account state	Makes balance and margin recovery queryable
`EventReplayResult`	Replay counts, offset, and projection	Separates recovery results from the event-store implementation
`EventReplayer`	Delivers persisted events to the synchronous bus	Reuses the runtime's existing dispatch semantics during recovery

5.5 Execution commands and order intents

Type	Usage	Design rationale
`SubmitOrder`	Rust-shaped command envelope	Carries order model, type, TIF, trigger, trailing, and emulation metadata
`CancelOrder`	Cancel command	Separates command from resulting canceled event
`ModifyOrder`	Fixed-point quantity, price, and trigger modification command	Mirrors Rust optional modify fields without integer truncation
`OrderType`	Market, limit, stop, and trailing variants	Explicit dispatch rather than implicit model inspection
`OrderIntent`	Market-style internal execution input	Lightweight transport from risk to venue
`LimitOrderIntent`	Limit-style internal execution input	Keeps limit price and trailing-limit data explicit
`SignalDirection`	`BUY`, `SELL`, `HOLD`	Shared side vocabulary for strategies, orders, and events
`TimeInForce`	`GTC`, `IOC`, `FOK`, `GTD`, `DAY`	Controls working-order lifetime and remainder handling
`TriggerType`	Trigger price source vocabulary	Matches Rust last, quote, mark, index, midpoint, and double-match concepts
`TrailingOffsetType`	`PRICE`, `BASIS_POINTS`, `TICKS`, `PRICE_TIER`	Java rejects `PRICE_TIER` because current Rust rejects it
`OrderAccepted`	Accepted market-style event	Makes venue acknowledgement observable
`LimitOrderAccepted`	Accepted limit-style event	Preserves order-type-specific event flow
`OrderDenied`	Risk/system denial event	Represents failure before venue acceptance
`LimitOrderDenied`	Limit risk denial event	Keeps limit denial payload typed
`OrderRejected`	Venue rejection event	Distinguishes venue rejection from local denial
`LimitOrderRejected`	Limit venue rejection event	Typed limit rejection counterpart
`OrderCanceled`	Successful cancel event	Drives terminal `CANCELED` transition
`OrderCancelRejected`	Failed cancel event	Restores the prior open state
`OrderModified`	Successful modify event	Makes accepted updates observable
`OrderModifyRejected`	Failed modify event	Restores the prior open state
`OrderExpired`	GTD/DAY expiry event	Drives terminal `EXPIRED` transition
`OrderTriggered`	Stop-limit trigger event	Drives `ACCEPTED -> TRIGGERED`

Type	Usage	Design rationale
`OrderEmulated`	Local emulator ownership event	Drives `INITIALIZED -> EMULATED`
`OrderReleased`	Emulator release event	Drives `EMULATED -> RELEASED`
`OrderVoided`	Fill-correction event	Represents the Rust terminal correction vocabulary
`OrderFill`	One execution fill	Carries quantity, price, commission, and liquidity side
`SettledOrderFill`	Fill after portfolio settlement	Separates raw venue fill from state-settled fill
`OrderState`	Immutable submitted/filled/remaining snapshot	Makes state queryable without exposing mutable order internals
`OrderStatus`	Full Rust status vocabulary	Predicates identify open and terminal states
`OrderStateMachine`	Validates every order transition	Prevents invalid lifecycle jumps and preserves prior pending state
`OrderMatchingEngine`	Stateless legacy single-price matcher	Retained as a small model-level utility; `SimulatedExchange` owns current book matching
`ExecutionClient`	Venue execution contract	Allows simulated and future live clients to share the boundary
`BacktestExecutionClient`	Adapts `SimulatedExchange` to `ExecutionClient`	Keeps execution-engine venue routing independent from simulator details
`ExecutionEngine`	Risk, emulator, client routing, state, and portfolio coordination	Central execution boundary corresponding to Rust execution orchestration
`SimulatedExchange`	Working orders, book consumption, triggers, TIF, latency, fills	Owns venue-specific matching and deterministic liquidity consumption
`OrderEmulator`	Holds locally emulated submit commands and releases on snapshots	Mirrors Rust emulator ownership before venue submission
`Commission`	Immutable fee amount/currency	Keeps fill fee data explicit and testable
`FeeModel`	Fee calculation contract	Supports interchangeable venue fee policies
`FixedFeeModel`	Fixed commission policy	Useful for per-order or one-time fees
`MakerTakerFeeModel`	Rate-based maker/taker policy	Calculates fee from notional and liquidity side
`PerContractFeeModel`	Quantity-based policy	Models contract-style fees without notional math
`ProbabilityPriceFeeModel`	Probability/price fee policy	Supports specialized instrument fee formulas
`CappedOptionFeeModel`	Capped option fee policy	Prevents fee growth beyond a configured cap
`TieredNotionalOptionFeeModel`	Notional-tier option fee policy	Models tiered fee schedules
`LiquiditySide`	`MAKER` or `TAKER`	Explicitly records how liquidity was consumed or provided
`LatencyModel`	Operation latency contract	Allows deterministic or future stochastic latency models
`StaticLatencyModel`	Base plus operation-specific latency	Current reproducible latency implementation
`VenueId`	Validated venue value object	Rust uses validated identity strings rather than enums

Type	Usage	Design rationale
`DeterministicOrderId`	Deterministic order identity helper	Supports reproducible event correlation
`ExecutionEngineConfig`	Immutable execution configuration DTO	Future configuration hook for execution policy

5.6 Order model package

Type	Usage	Design rationale
`Order`	Sealed common order interface	Gives factories and submit commands one typed model boundary
`MarketOrder`	Immediate market order model	Immutable validated order metadata
`LimitOrder`	Resting/crossing limit order model	Separates limit price from market execution
`StopMarketOrder`	Triggered market order model	Carries trigger price and trigger source
`StopLimitOrder`	Triggered limit order model	Carries both trigger and limit price
`TrailingStopMarketOrder`	Dynamic stop-market model	Carries activation and offset metadata
`TrailingStopLimitOrder`	Dynamic stop-limit model	Adds limit offset to trailing stop metadata
`Money`	Immutable amount/currency value	Exact `BigDecimal` amount with currency-aware rounding and a legacy double projection

5.7 Indicators package

Type	Usage	Design rationale
`ExponentialMovingAverage`	Stateful EMA updates	Keeps indicator state in Java and minimizes bridge calls
`SimpleMovingAverage`	Stateful rolling average	Matches strategy warm-up behavior with explicit count/initialization

5.8 Message bus package

Type	Usage	Design rationale
`MessageBus`	Typed synchronous publish/send/request/response	Core in-memory runtime path; no queue by default
`TypedTopicRouter`	Wildcard topic matching with cached indexes	Separates topic routing from generic bus storage
`TypedEndpointMap`	Exact typed endpoint lookup	Models point-to-point Rust endpoint delivery
`Handler`	Generic typed callback	Keeps message bus API type-safe
`Serializer`	Serialization contract	Boundary for external bus messages
`JacksonSerializer`	Jackson implementation	Uses existing project dependency for structured payloads
`BusMessage`	External message envelope	Carries payload type and serialized payload
`MessageBusEvent`	Topic/payload event record	Data unit for transport-backed bus implementations
`MessageBusRouter`	Legacy untyped routing contract	Extension point for non-generic bus consumers

Type	Usage	Design rationale
`MessageHandler`	Legacy untyped callback contract	Supports the Disruptor-shaped scaffold
`MessageBusBacking`	External transport backing contract	Separates transport buffering from synchronous bus semantics
`RingBufferMessageBusBacking`	Bounded `ArrayBlockingQueue` backing	External-flow buffering with explicit full-buffer behavior
`DisruptorMessageBus`	LMAX Disruptor-shaped facade	Performance experiment/scaffold; publish integration remains TODO
`SerializationEncoding`	Encoding vocabulary	Configuration marker for serialized messages

5.9 Portfolio, risk, and reconciliation packages

Type	Usage	Design rationale
`Portfolio`	Applies fills, updates positions, tracks average price, PnL, and account settlement	Keeps position and account transitions coordinated after execution
`PositionUpdate`	Post-fill position/PnL event	Makes portfolio transition observable
`PortfolioConfig`	PnL/snapshot options	Configuration boundary for future account behavior
`AccountLedger`	Tracks currency balances, order reservations, position margin, FX conversion, and settlement	Isolates account mutation from order matching and supports deterministic snapshots
`AccountState`	Immutable primary balance and margin snapshot with per-currency balances	Mirrors Nautilus `AccountState` as an observable accounting boundary
`AccountBalance`	Total, locked, and free amount for one currency	Preserves the invariant `total = locked + free`
`AccountType`	`CASH` or `MARGIN` settlement policy	Makes borrowing and position-reservation semantics explicit
`AccountStateEvent`	Publishes an account snapshot after acceptance or settlement	Allows account changes to enter the same event evidence stream
`AccountMarginCall`	Typed notification when equity falls below maintenance margin	Separates a warning threshold from ordinary account snapshots
`AccountLiquidationRequired`	Typed notification when equity cannot support maintenance	Makes liquidation-required state observable without silently inventing a fill
`RiskEngine`	Validates side, quantity, instrument, price, notional, trading state, and available margin	Keeps risk before venue execution and prevents unsupported exposure
`RiskDecision`	Allow/reject result	Separates risk result from order event emission
`RiskEngineConfig`	Immutable risk options	Future policy configuration boundary
`TradingState`	Active, halted, reducing state vocabulary	Enables global risk policy changes
`ReconciliationComparator`	Java-side comparison utility	Keeps semantic comparison logic available inside the Java module
`ReconciliationResult`	Immutable comparison result	Carries match status, row count, and mismatch information

5.10 System and trading packages

Type	Usage	Design rationale
`NautilusKernel`	Composition root and chronological replay owner	Mirrors Nautilus runtime orchestration and owns global input sequence
`NautilusKernelBuilder`	Kernel construction helper	Provides a future fluent composition boundary
`NautilusKernelConfig`	Kernel options	Separates runtime policy from kernel implementation
`Clock`	Clock abstraction	Allows simulated and future real clocks
`SimulatedClock`	Replay clock	Makes event time explicit and testable
`ComponentLifecycle`	Shared lifecycle transition table	Centralizes valid component state changes
`ComponentState`	Lifecycle state vocabulary	Mirrors initialized/ready/running/stopped/faulted concepts
`ComponentTrigger`	Lifecycle transition vocabulary	Keeps trigger names separate from state names
`Trader`	Strategy registration, contexts, and bar subscriptions	Owns strategy lifecycle and symbol routing
`Actor`	Actor lifecycle contract	Structural boundary for event-driven components
`StrategyHandler`	Java callback contract used by Python proxy	Small callback surface for strategy execution
`StrategyContext`	Current bar, position, sequence, and order API	Gives strategies controlled runtime access
`OrderApi`	Creates and publishes typed orders; cancel/modify/emulation	Keeps strategy code independent from command construction details
`StrategySignal`	Strategy-generated signal event	Separates intent observation from order submission
`ExecutionAlgorithm`	Execution-algorithm contract	Extension point for TWAP/VWAP-like algorithms

6. Important Runtime Details

6.1 Order lifecycle

```
INITIALIZED
    |
    +--> DENIED
    |
    +--> EMULATED --> RELEASED --> SUBMITTED --> ACCEPTED
                                       |             |
                                       |             +--> PENDING_UPDATE --> ACCEPTED
                                       |             |                    \-> PARTIALLY_FILLED
                                       |             +--> PENDING_CANCEL --> CANCELED
                                       |             +--> REJECTED
                                       |
                                       +--> PARTIALLY_FILLED --> FILLED
                                       |                    \-> EXPIRED / CANCELED
                                       +--> EXPIRED
                                       +--> CANCELED
                                       +--> FILLED
                                       +--> VOIDED after fill correction
```

Stop-market orders execute immediately on trigger. Stop-limit orders emit `TRIGGERED` and then match as limit orders. Trailing orders first activate and ratchet their trigger/limit prices before they can trigger.

6.2 Order-book behavior

`OrderBookSnapshot` replaces the visible L2 book. `OrderBookDelta` mutates a persistent book:

```
ADD     -> add quantity at price
UPDATE  -> replace quantity at price
DELETE  -> remove price level
CLEAR   -> clear one side
```

Market buys read asks from lowest to highest. Market sells read bids from highest to lowest. Crossed limits take available opposite-side liquidity; unfilled limits rest as makers. Working strategy orders are sorted by better price first and insertion sequence second.

For L3 MBO, each `VenueOrder` remains distinct. Orders at the same price are ordered by venue sequence and then ID. A resting client order snapshots the quantity and IDs ahead of it. Venue deletes and size decreases advance that queue; size increases retain the venue order's position. A trade tick advances only the passive side selected by its aggressor side, consuming queue-ahead quantity before a client maker order receives a fill.

6.3 Account and margin behavior

`AccountLedger` is deliberately downstream of execution and upstream of account-state logging:

```
accepted order -> reserve initial margin or cash
venue fill     -> release proportional reservation
               -> apply commission and realized PnL
               -> update position margin
               -> publish AccountStateEvent
cancel/expiry/reject -> release reservation
```

Margin accounts reserve quote-currency notional multiplied by the instrument's initial-margin rate and divided by leverage. Open positions report both initial and maintenance margin using `InstrumentSpec`. Cash accounts reserve buy-side funds, settle buy/sell notional in the quote currency, and reject uncovered sells. Additional currencies can be deposited and converted through explicitly configured FX rates; missing conversion data rejects a reservation rather than silently treating currencies as equal.

These rules are a deterministic backtest baseline. They do not yet model every Nautilus derivative margin model, liquidation execution policy, or live FX feed.

6.4 Numeric representation

Quantities use immutable fixed-point `Quantity` values backed by a `long` raw value and explicit decimal precision. `InstrumentSpec` validates both the configured `sizePrecision`/`sizeIncrement` and `pricePrecision`/`priceTickSize`. Binance `LOT_SIZE.stepSize`, `minQty`, and `PRICE_FILTER.tickSize` are mapped into the same contract and invalid orders are rejected before REST or simulated execution. Python accepts integers, `Decimal`, and decimal strings for exact quantities and tick metadata. Monetary values use exact `BigDecimal` storage through commissions, balances, FX conversions, reservations, margins, and equity; `CurrencyPrecision` provides explicit currency rounding at external boundaries. Python exposes account values as `Decimal`. Prices and PnL at the execution boundary remain primitive `double` where the existing matching API requires them, while configured order-price validation uses exact `Decimal`/`BigDecimal` conversion before dispatch.

7. Python Integration

The Python facade is intentionally thin:

```
abc_trading.backtest.engine.BacktestEngine
        |
        +--> starts JVM and loads Java classes
       +--> converts Python Bars/Snapshots and exact decimal quantities to Java objects
        +--> installs JProxy StrategyHandler
        +--> exposes market, limit, stop, trailing, cancel, modify
```

Python strategy callbacks receive a Java-owned `StrategyContext` wrapper. Java increments the global input sequence before publishing each input, so Python-generated signals and Java-generated execution events share the same chronology.

7.1 Live adapter flow

```
Binance public WebSocket
       |
       v
BinanceMessageMapper -> BinanceFuturesLiveRuntime -> MessageBus
                                                |
                      +-------------------------------+----------------+
                      |                                                |
                      v                                                v
               Strategy/Portfolio                                ExecutionEngine
                                                                  |
                                                                  v
                                                Binance signed REST + user stream
```

`NautilusKernel.addBinanceFutures` registers the runtime as both a data client and an execution client. Kernel lifecycle starts/stops the adapter; public depth, trades, and mark prices enter the core bus, while user-data fills become `OrderFill` events and account snapshots remain typed adapter/account events. The adapter uses JDK `HttpClient` and WebSocket APIs, so no third-party live transport is required.

8. Reconciliation Evidence

Existing bar workflow

```
shared immutable OHLCV CSV
        |
        +--> Java BacktestEngine
        |
        `--> Nautilus BacktestEngine
                 |
                 `--> semantic CSV comparator
```

Compared semantic events:

- `SIGNAL`
- `ORDER_FILL`
- `POSITION_UPDATE`

Backend-only transport events and backend-specific identifiers are excluded intentionally. Prices, PnL, and commissions are compared numerically.

Order-book workflow

Shared fixture:

```
recon/order_book_market_data.csv
```

Both implementations produce:

```
market-buy-1  101.00 x 3  TAKER
market-buy-1  101.01 x 3  TAKER
market-sell-1  99.00 x 5  TAKER
market-sell-1  98.99 x 2  TAKER
limit-buy-1   100.00 x 2  MAKER
```

The order-book comparator checks order, price, quantity, and liquidity side for every fill.

Account and persistence evidence

The account fixture compares eight final fields:

```
MATCH account state fields=8
```

The event-store integration writes the same canonical events to CSV and versioned JSONL, then replays them into a synchronous bus and state projection. Quantities, signed positions, realized PnL, commissions, balances, margins, and equity serialize as canonical decimal strings and are reconstructed without a `double` conversion. The persistence suite includes high-precision JSONL and CSV round-trip coverage. The Binance adapter contract suite validates URL routing, HMAC signing, wire mapping, decimal preservation, REST order parameters, and the kernel bridge without requiring credentials.

L3 MBO workflow

Shared fixture:

```
recon/l3_mbo_market_data.json
```

The Java runner and Nautilus runner both load the same individual venue orders, submit the same limit and market orders, process the same seller-aggressor trades, and emit compact fill rows. The comparator

verifies client order, price, quantity, liquidity side, and fill order:

```
MATCH L3 fills=2
```

Nautilus `venue_order_id` identifies the client order's venue assignment. The Java simulator additionally records the passive L3 book order ID when it is known; that diagnostic identity is intentionally not treated as a Rust parity field because it is not present in Nautilus `OrderFilled`.

Account workflow

Account evidence is emitted through the same Java CSV stream as lifecycle events. `ACCOUNT_STATE` rows contain the primary currency, total, locked and free balances, initial margin, and maintenance margin. The account tests cover margin reservations, cash notional settlement, commissions, realized PnL, instrument-specific rates, FX conversion, uncovered cash sells, and additional currency balances.

9. Tests and Validation

Current Java tests cover:

- message bus routing and ordering
- lifecycle transitions
- kernel sorting and clock behavior
- risk validation
- order API construction
- fee and latency models
- order state transitions
- partial and multiple fills
- cancel, modify, expiry, IOC, FOK, and GTD
- stop and trailing triggers
- emulation release
- tick configuration
- L2 depth traversal and FIFO
- persistent order-book deltas
- individual L3 venue-order identity and queue-ahead changes
- trade-driven queue consumption and aggressor-side behavior
- portfolio position/PnL behavior
- cash/margin account settlement and account-state events
- account-state parity across balance, margin, unrealized PnL, and equity
- persistent JSONL event storage, replay projections, and checkpoint recovery
- Binance adapter URL, signing, market-data, user-data, and kernel integration

The Java module currently contains 168 production source files and 20 test files. The test suite is intentionally focused on the current deterministic backtest and one live-adapter slice; it is not yet a complete replacement of all Nautilus modules.

10. Current Gaps and Recommended Next Work

1. **Exact account parity:** broader Rust account-event fixtures, full derivative margin models, and richer market-driven FX coverage remain.
2. **Binance live hardening:** authenticated Testnet order-flow smoke tests, exchange-info filter enforcement, WebSocket sequence-gap recovery, and user-stream account reconciliation.
3. **Additional adapters:** add other venues only after the Binance contract and operational path are stable.
4. **Broader account parity:** extend exact monetary parity to additional Rust account-event and derivative-margin fixtures.
5. **Execution algorithms:** actual algorithm scheduling rather than the current interface boundary.

6. **Disruptor integration:** complete the `DisruptorMessageBus` publication path only if profiling shows the synchronous bus is insufficient.

11. Design Conclusion

The Java implementation has moved beyond isolated skeleton classes. It now forms a coherent deterministic backtest runtime with explicit Rust-shaped ownership boundaries, durable replay evidence, and one Binance USD-M live-adapter slice. The next work is operational Testnet hardening and deeper account parity, not multiplying adapters before the first one is proven.

12. Business Knowledge and Rust Reference

12.1 Why these boundaries matter to a trading system

The rewrite follows trading-domain ownership rather than treating the application as a generic event processor:

Business concept	Operational meaning	Java owner	Nautilus/Rust reference
Market data	The venue's observable state and executed trades	`DataEngine`, `OrderBook*`, `TradeTick`	`crates/data`, `crates/model/src/data`
Price-time priority	Earlier and better-priced liquidity receives execution first	`SimulatedExchange.L3BookState`	`crates/model/src/orderbook`, matching engine
Queue ahead	A passive order cannot fill until visible liquidity ahead is consumed or removed	`WorkingOrder` queue fields and L3 state	`crates/execution/src/matching_engine/engine.rs` queue methods
Maker/taker	Determines execution role and often fee schedule	`LiquiditySide`, `FeeModel`	`LiquiditySide`, fill-model paths in matching engine
Risk	Prevents invalid or unaffordable exposure before routing	`RiskEngine`	`crates/risk/src/engine`
Free versus locked funds	Separates immediately tradable capital from order/position commitments	`AccountLedger`, `AccountState`	`AccountBalance`, `CashAccount`, `MarginAccount`

Business concept	Operational meaning	Java owner	Nautilus/Rust reference
Initial versus maintenance margin	Initial margin gates new exposure; maintenance margin measures ongoing safety	`InstrumentSpec`, `AccountLedger`	`MarginBalance`, `AccountsManager`
Realized PnL	PnL becomes final when a position is reduced or closed	`Portfolio`	`AccountsManager::update_balances`, position PnL methods
Settlement	A fill changes positions, balances, commissions, and observable state	`ExecutionEngine` and `Portfolio`	portfolio event and account-state flow

12.2 Rust source map

The Java implementation is anchored to these Nautilus areas:

Java area	Nautilus reference	Reason for the mapping
`MessageBus` and topic routing	`crates/common/src/message_bus` and component messaging	Preserve synchronous in-process dispatch semantics for deterministic replay
`NautilusKernel` and lifecycle	`crates/system`, `crates/common/src/actor`	Keep composition, clock, lifecycle, and ownership explicit
Order models and state machine	`crates/model/src/orders`, `crates/execution/src/engine`	Match validated order vocabulary and legal transitions
`SimulatedExchange`	`crates/backtest/src/exchange`, `crates/execution/src/matching_engine`	Keep venue matching, triggers, TIF, fills, and queue behavior together
L3 queue tracking	`engine.rs`: `snapshot_queue_position`, `decrement_queue_on_trade`, `advance_l3_queue_on_delete`, `adjust_l3_queue_on_update`	Preserve quantity-ahead and per-book-order behavior
`Portfolio` and `AccountLedger`	`crates/portfolio/src/portfolio.rs`, `manager.rs`, `accounts/cash.rs`, `accounts/margin.rs`	Separate positions, balance settlement, and margin policy
`RiskEngine`	`crates/risk/src/engine/mod.rs`	Perform affordability and exposure checks before execution
CSV reconciliation	`crates/model` events and Python backtest surfaces	Compare observable semantics while excluding backend-only IDs

12.3 Business interpretation of a fill

For a market buy, the simulator consumes the lowest available asks. For a resting buy, a seller-aggressor trade first consumes the bid queue ahead and only the excess can fill the client. A fill then has four consequences:

1. The order state advances by the fill quantity.
2. The portfolio position and average price change.

3. Commission and realized PnL change account equity.
4. Reserved funds and position margin are recalculated and published.

This sequence is why matching, risk, portfolio, and account settlement are separate classes even though one market event can touch all of them.